Part 2: Descriptive Statistics

Understand the retail business before making a decision · 307312 Applied Statistics

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- **Management question:** Is a strong sales total hiding weak profitability?
- **Business:** Superstore sells Furniture, Office Supplies and Technology.
- **Scope:** US orders placed in 2023–2025; Tableau's fictitious sample company.
- **Methods:** Tables, charts, centre, spread, association and distribution shape.

2.1 From a question to evidence

1. State the decision and the population or records of interest.
2. Identify the observational unit and relevant variables.
3. Check definitions, missingness and repeated records.
4. Summarize with a table, chart and numerical measures.
5. Explain the finding, limitations and next investigation.

- **Descriptive statistics:** summarize the records you have.
- **Inferential statistics:** use a sample or model to reason beyond observed records.
- A large dataset does not automatically represent all retailers, customers or future years.

Import and establish the unit

```
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
knitr::kable(data.frame(unit = c("Product line", "Order", "Customer"),
  records = c(nrow(lines), nrow(orders), n_distinct(lines$customer_id))))
```

unit	records
Product line	6682
Order	3322
Customer	782

- **6,682 lines:** product entries within orders.
- **3,322 orders:** aggregate sales and profit by `order_id`.
- **782 customers:** some place more than one order.
- A category can occur several times within an order; category line counts are not counts of customers.

2.2 Variables and measurement

Variable	Business meaning	Statistical treatment
`order_id`, `customer_id`, `postal_code`	Identify records, customers or places	Labels; arithmetic is meaningless
`category`, `region`, `segment`, `ship_mode`	Product, market and shipping classifications	Nominal categorical
`quantity`	Units sold on a line	Discrete numerical, ratio scale
`sales`	Recorded line sales in dollars	Numerical, ratio scale
`profit`	Recorded profit, including negative amounts	Numerical; avoid ratios of profits with arbitrary/negative bases
`discount`	Fractional discount, e.g. 0.20	Numerical; multiply by 100 for percent
`order_date`	Calendar date	Time; differences are meaningful
`ship_days`	Ship date minus order date	Discrete duration in days

- **Nominal:** categories with no required ranking.
- **Ordinal:** ordered categories without established equal spacing, such as satisfaction ratings.
- **Interval:** equal differences, without an absolute zero, such as Celsius temperature.
- **Ratio:** meaningful zero and ratios, such as units or elapsed time.
- Superstore's ship modes are service labels; the course does not assume an equal numerical spacing between them.

Sources and collection

- Operational records capture transactions; surveys capture responses; experiments assign conditions.
- A census covers every member of a defined frame; a sample covers a subset.
- Probability sampling supports design-based inference when the sampling mechanism is known.
- Convenience samples, nonresponse and coverage gaps may bias conclusions.
- Superstore is fictitious. Its complete course extract is a **finite teaching frame**, not a sample of the real retail industry.

Check data before summarizing

```
stopifnot(!anyDuplicated(lines$row_id), all(lines$sales > 0),
          all(lines$discount >= 0 & lines$discount <= 1),
          all(orders$ship_days >= 0))
colSums(is.na(lines))
```

```
      row_id      order_id    order_date     ship_date     ship_mode
           0             0             0             0             0
 customer_id customer_name       segment       country          city
           0             0             0             0             0
       state   postal_code        region    product_id      category
           0             0             0             0             0
sub_category  product_name         sales      quantity      discount
           0             0             0             0             0
      profit return_status
           0             0
```

```
range(as.Date(lines$order_date))
```

```
[1] "2023-01-03" "2025-12-31"
```

- Repeated order IDs are expected; repeated line IDs would require investigation.
- Postal codes stay as text.
- The course does not delete genuine negative profits or large sales values.

2.3 Frequency tables and percentages

Question: What does the product mix look like?

```
category_mix <- lines %>% count(category, name = "line_count") %>%
  mutate(percent = 100 * line_count / sum(line_count))
knitr::kable(category_mix, digits = 1)
```

category	line_count	percent
Furniture	1435	21.5
Office Supplies	4024	60.2
Technology	1223	18.3

- Frequency: number of observations in a category.
- Relative frequency: category frequency divided by the total.
- Here, the denominator is order lines, not orders, units or revenue.
- Changing the measure can change the apparent importance of a category.

Joint and conditional percentages

Question: Does category mix differ by customer segment?

```
segment_category <- xtabs(~ segment + category, data = lines)
knitr::kable(segment_category)
```

	Furniture	Office Supplies	Technology
Consumer	761	2106	656
Corporate	448	1229	363
Home Office	226	689	204

```
knitr::kable(100 * prop.table(segment_category, margin = 1), digits = 1)
```

	Furniture	Office Supplies	Technology
Consumer	21.6	59.8	18.6
Corporate	22.0	60.2	17.8
Home Office	20.2	61.6	18.2

- The first table contains counts.
- `prop.table(..., 1)` divides each cell by its **row total**.
- Each row therefore totals 100%: category mix **within a segment**.
- `prop.table(segment_category)` instead uses all 6,682 lines as the denominator.

- Differences between percentages are **percentage points**.

An ordered cumulative table

Question: What proportion of orders ship within each number of days?

```
shipping <- orders %>% count(ship_days, name = "orders") %>%
  arrange(ship_days) %>%
  mutate(percent = 100 * orders / sum(orders), cumulative_pct = cumsum(percent))
knitr::kable(shipping, digits = 1)
```

ship_days	orders	percent	cumulative_pct
0	168	5.1	5.1
1	111	3.3	8.4
2	456	13.7	22.1
3	319	9.6	31.7
4	935	28.1	59.9
5	741	22.3	82.2
6	396	11.9	94.1
7	196	5.9	100.0

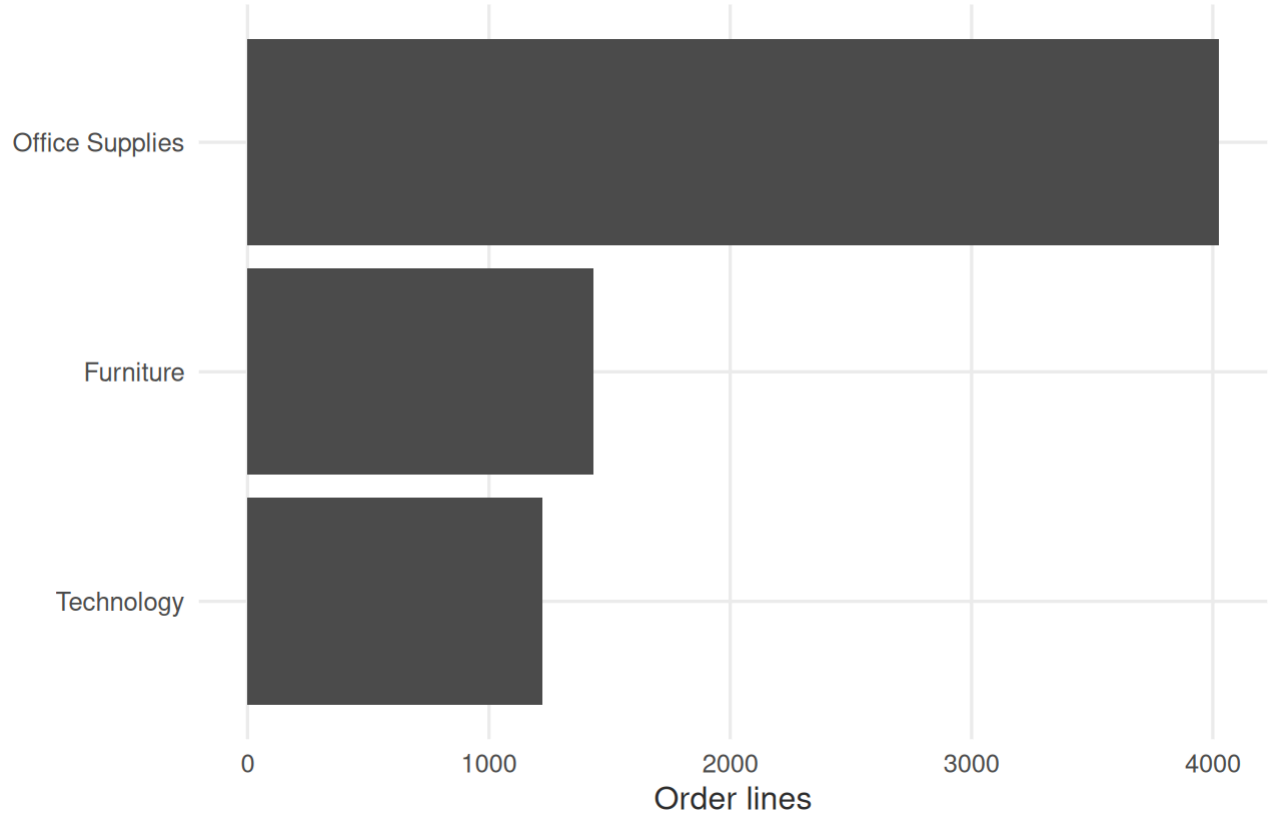
- Accumulation makes sense for an ordered duration.
- It does not add useful meaning to an arbitrary alphabetical list of categories.
- Shipping lag ends at dispatch, not arrival at the customer's address.

2.4 Choose a chart for the question

Question	Chart	What to check
Which categories are most common?	Bar chart	Counts or percentages; start quantitative bar axes at zero
How does product mix differ by segment?	Grouped or 100% stacked bars	Consistent denominators and category mapping
How large are typical orders?	Histogram	Units, bin width and long tails
How do group distributions differ?	Side-by-side boxplots	Sample sizes, medians, spread and unusual points
Do sales and profit move together?	Scatterplot	Unit of observation, outliers and nonlinear patterns
How does revenue change over time?	Time-series line chart	Time order, incomplete periods and seasonality

Category counts

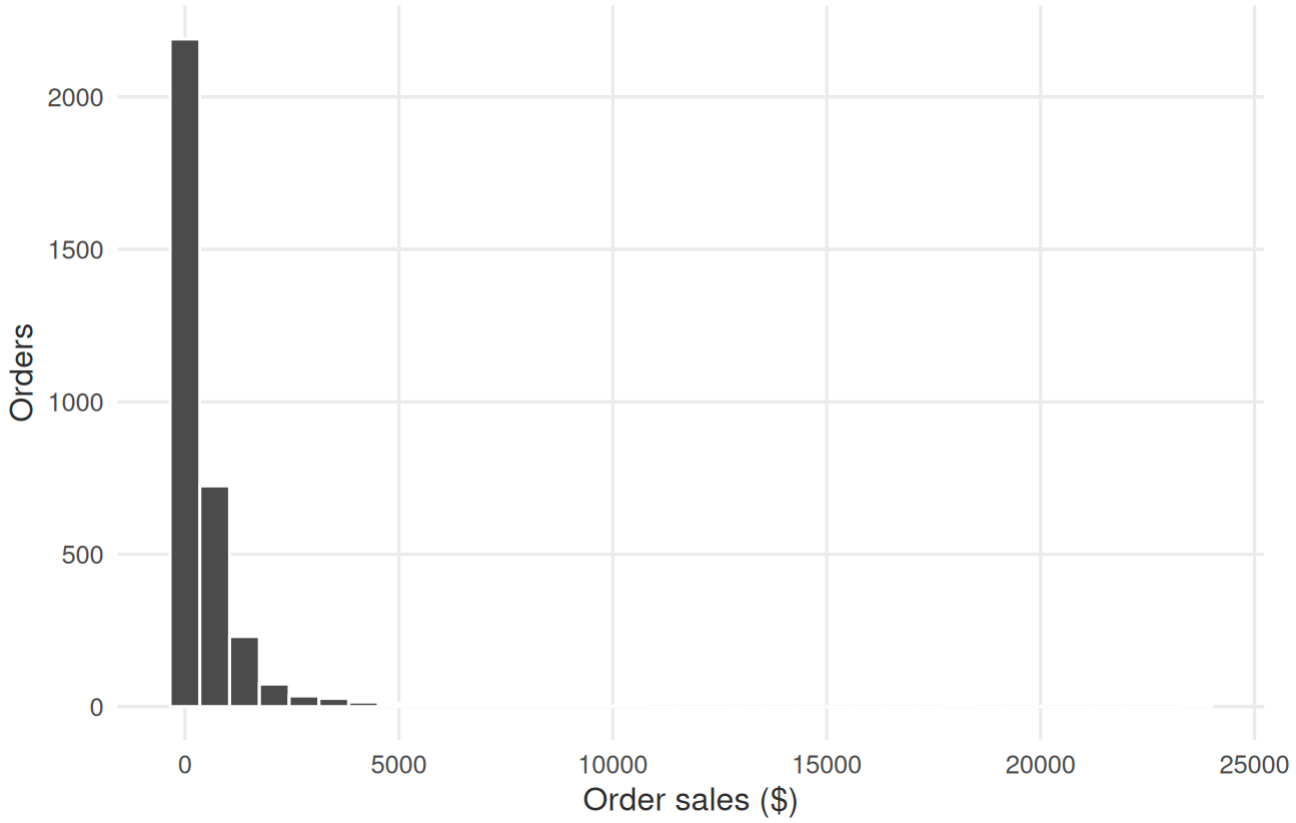
```
ggplot(category_mix, aes(reorder(category, line_count), line_count)) +
  geom_col(fill = petra_ink) + coord_flip() +
  labs(x = NULL, y = "Order lines")
```



Office Supplies dominates the count of product lines; this is not a revenue ranking.

A histogram of order values

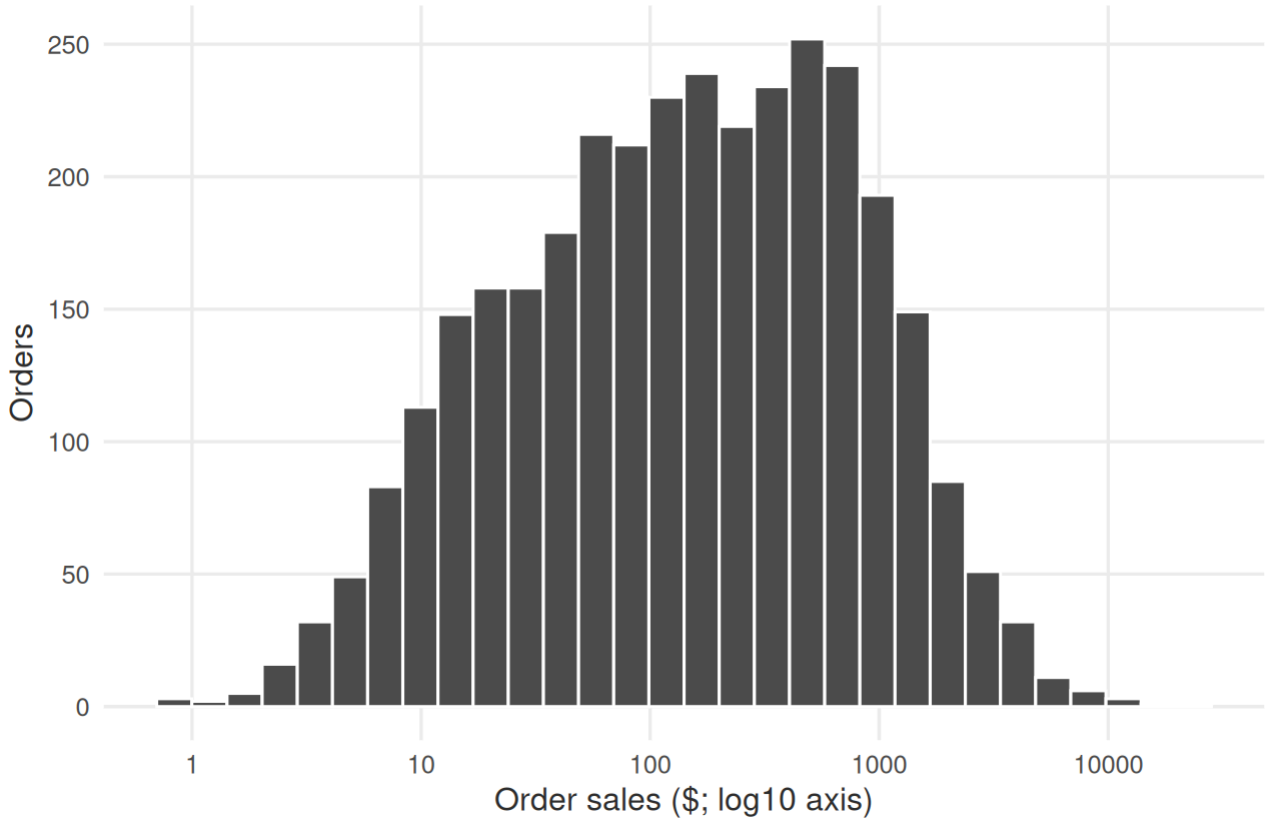
```
ggplot(orders, aes(sales)) +
  geom_histogram(bins = 35, fill = petra_ink, colour = "white") +
  labs(x = "Order sales ($)", y = "Orders")
```



Order values are concentrated at the lower end with a long right tail; all orders are retained.

- Adjacent bars represent numerical intervals.
- Compare at least two bin widths before deciding a pattern is stable.
- A log-scaled companion can reveal detail among small orders; label the transformation clearly.

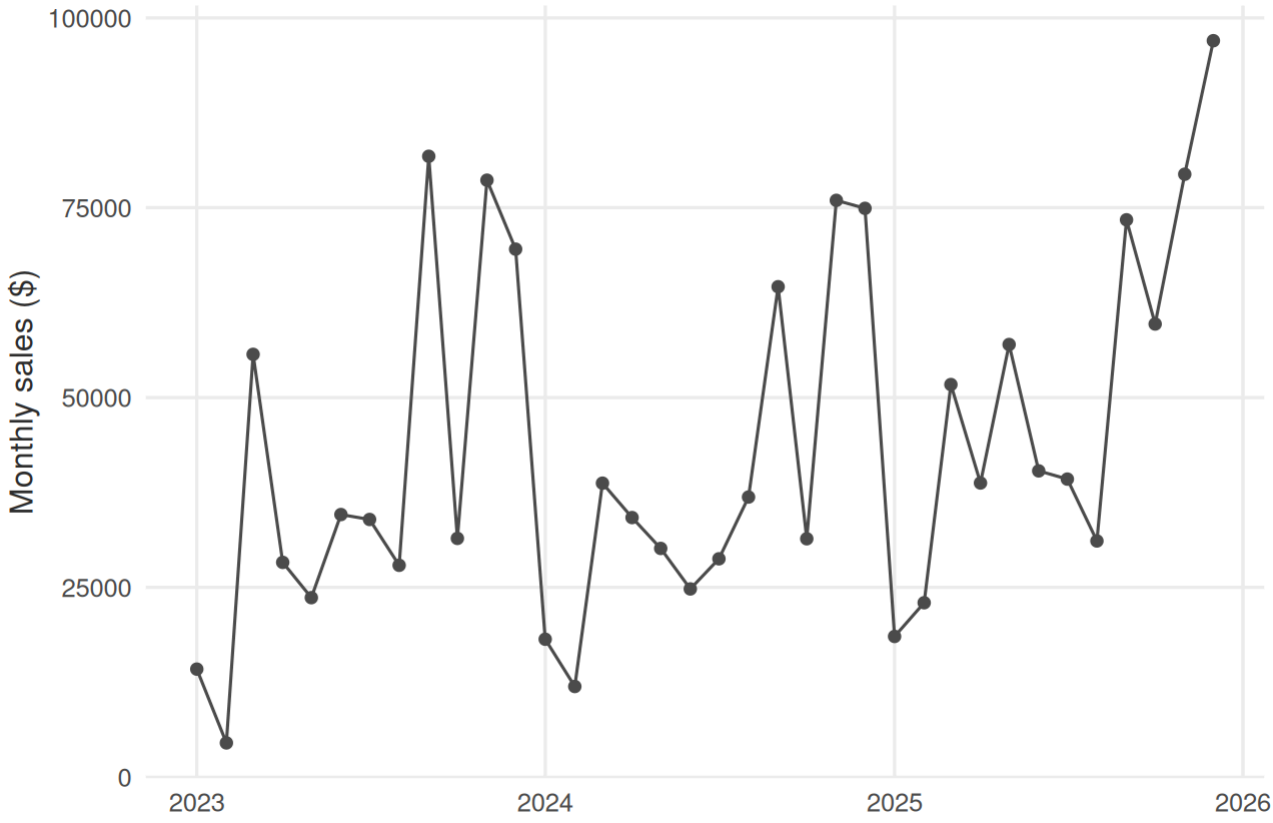
```
ggplot(orders, aes(sales)) +
  geom_histogram(bins = 30, fill = petra_ink, colour = "white") +
  scale_x_log10() + labs(x = "Order sales ($; log10 axis)", y = "Orders")
```



The same orders on a logarithmic sales axis; equal distances represent equal ratios.

Compare monthly totals

```
monthly <- orders %>% mutate(month = as.Date(format(order_date, "%Y-%m-01"))) %>%
  group_by(month) %>% summarise(total_sales = sum(sales), .groups = "drop")
ggplot(monthly, aes(month, total_sales)) +
  geom_line(colour = petra_ink) + geom_point(colour = petra_ink) +
  labs(x = NULL, y = "Monthly sales ($)")
```



- Monthly totals combine order frequency and order size.
- An upward or seasonal pattern does not identify a causal driver.
- We use three complete order years; some late-December orders ship in the next year.

2.5 Measures of central location

For observations $x_1, \ldots, x_n$, the arithmetic mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i .$$

- **Mean:** total divided by count; affected by unusually large values.
- **Median:** middle ordered value, or the mean of the two middle values.
- **Mode:** most frequent value/category; ties are possible.
- For continuous amounts, exact-value modes often reflect rounding or repeated prices.

```
centre <- orders %>% summarise(orders = n(), mean_sales = mean(sales),
                               median_sales = median(sales), total_sales = sum(sales))
knitr::kable(centre, digits = 2)
```

orders	mean_sales	median_sales	total_sales
3322	470.8	153.98	1563986

- Mean order sales are about **$471**; the median is about **$154**.
- The gap is consistent with the long right tail.
- Use the median to describe a typical order; use the mean with the order count to explain revenue.
- Neither measure alone shows the full distribution.

Counts, totals and weighted averages

```
category_performance <- lines %>% group_by(category) %>%
  summarise(line_count = n(), units = sum(quantity), total_sales = sum(sales),
            total_profit = sum(profit), .groups = "drop") %>%
  mutate(margin_pct = 100 * total_profit / total_sales)
knitr::kable(category_performance, digits = 2)
```

category	line_count	units	total_sales	total_profit	margin_pct
Furniture	1435	5591	526612.5	15432.88	2.93
Office Supplies	4024	15230	472949.9	82754.18	17.50
Technology	1223	4576	564423.2	94770.69	16.79

- Office Supplies has the most lines, but Technology has the largest sales total here.
- Furniture records a much thinner overall margin.
- Overall margin is a sales-weighted mean of line margins:

$$\frac{\sum_i \text{profit}_i}{\sum_i \text{sales}_i} = \sum_i \left(\frac{\text{sales}_i}{\sum_j \text{sales}_j}\right)\left(\frac{\text{profit}_i}{\text{sales}_i}\right).$$

- Do not average category percentages without considering the weights.
- Do not describe `sales / quantity` as an undiscounted list price.

2.6 Spread and variation

- **Range:** maximum minus minimum; depends only on two observations.
- **Mean absolute deviation from the mean:** average absolute distance from the mean.
- **Sample variance:** $s^2 = \sum (x_i - \bar{x})^2/(n-1)$.
- **Sample standard deviation:** $s = \sqrt{s^2}$, in the original units.
- **Frame variance:** divide by N when describing the entire fixed frame as a population.
- **IQR:** third quartile minus first quartile; describes the middle 50%.

```
x <- orders$sales
spread <- data.frame(
  measure = c("Range", "Mean absolute deviation", "Sample SD (n-1)",
              "Frame SD (N)", "IQR"),
  dollars = c(diff(range(x)), mean(abs(x - mean(x))), sd(x),
              sqrt(mean((x - mean(x))^2)), IQR(x)))
knitr::kable(spread, digits = 2)
```

measure	dollars
Range	23660.39
Mean absolute deviation	501.68
Sample SD (n-1)	982.97
Frame SD (N)	982.83
IQR	496.62

- R's `var()` and `sd()` use $n-1$.
- Variance has squared units; SD has dollar units here.

- R's `mad()` is a scaled **median** absolute deviation, not the mean absolute deviation above.
- Variation across orders is not variation over time for an individual customer.

Coefficient of variation

$$CV = 100\frac{s}{\bar{x}}.$$

```
100 * sd(orders$sales) / mean(orders$sales)
```

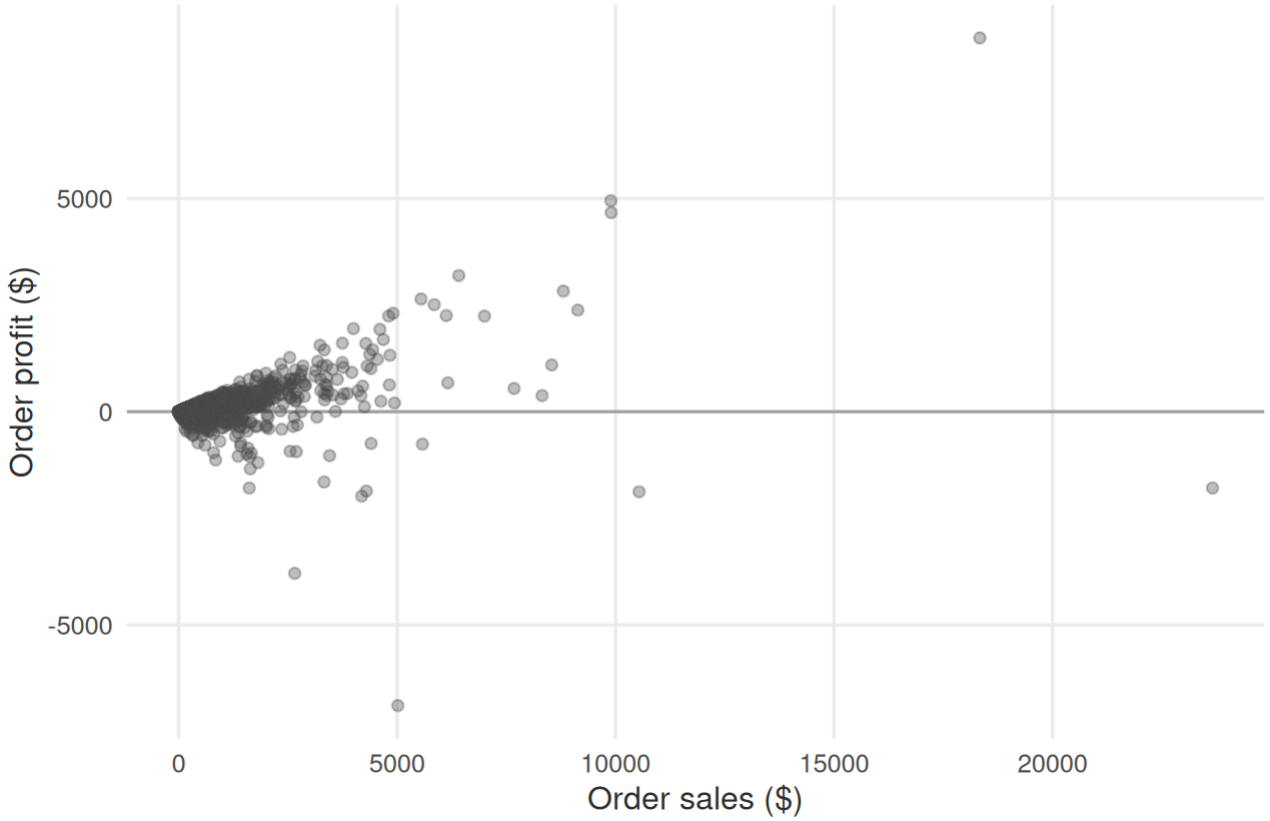
```
[1] 208.7897
```

- CV expresses SD relative to a positive mean on a meaningful ratio scale.
- It can compare relative variation across positive measures with different scales.
- Avoid CV for profit or margins that can be negative or have a mean near zero.
- A high CV is descriptive; it does not by itself measure business risk.

2.7 Association between numerical variables

Question: Are larger orders necessarily more profitable?

```
ggplot(orders, aes(sales, profit)) +
  geom_hline(yintercept = 0, colour = "#A3A3A3") +
  geom_point(alpha = 0.35, colour = petra_ink) +
  labs(x = "Order sales ($)", y = "Order profit ($)")
```



Order-level sales and profit; large orders can also record large losses.

Covariance and Pearson correlation

$$s_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n - 1}, \qquad r = \frac{s_{xy}}{s_x s_y}.$$

```
c(covariance = cov(orders$sales, orders$profit),
  pearson_r = cor(orders$sales, orders$profit),
  spearman_r = cor(orders$sales, orders$profit, method = "spearman"))
```

```
  covariance    pearson_r   spearman_r
1.509486e+05 4.504704e-01 5.108097e-01
```

- Covariance has product units; its magnitude changes when units change.
- Pearson r is unitless and lies between −1 and +1 for nonconstant variables.
- Pearson measures linear association; zero correlation does not imply no relationship.
- Spearman correlation describes rank association and is a useful companion for skewed data.
- Both can conceal group differences. Neither establishes causation.
- Profit and sales are related accounting measures; a correlation is not an independent causal explanation.

2.8 Distribution shape, quartiles and unusual observations

- **Right-skewed:** long right tail; mean often exceeds median.
- **Left-skewed:** long left tail; mean often falls below median.
- Symmetry alone does not establish a normal distribution.
- A numerical skewness index is useful alongside a histogram, not as a substitute.

```
c(mean = mean(x), median = median(x),
  pearson_second_skewness = 3 * (mean(x) - median(x)) / sd(x))
```

```
                mean                median pearson_second_skewness
           470.7963893           153.9820000                0.9669055
```

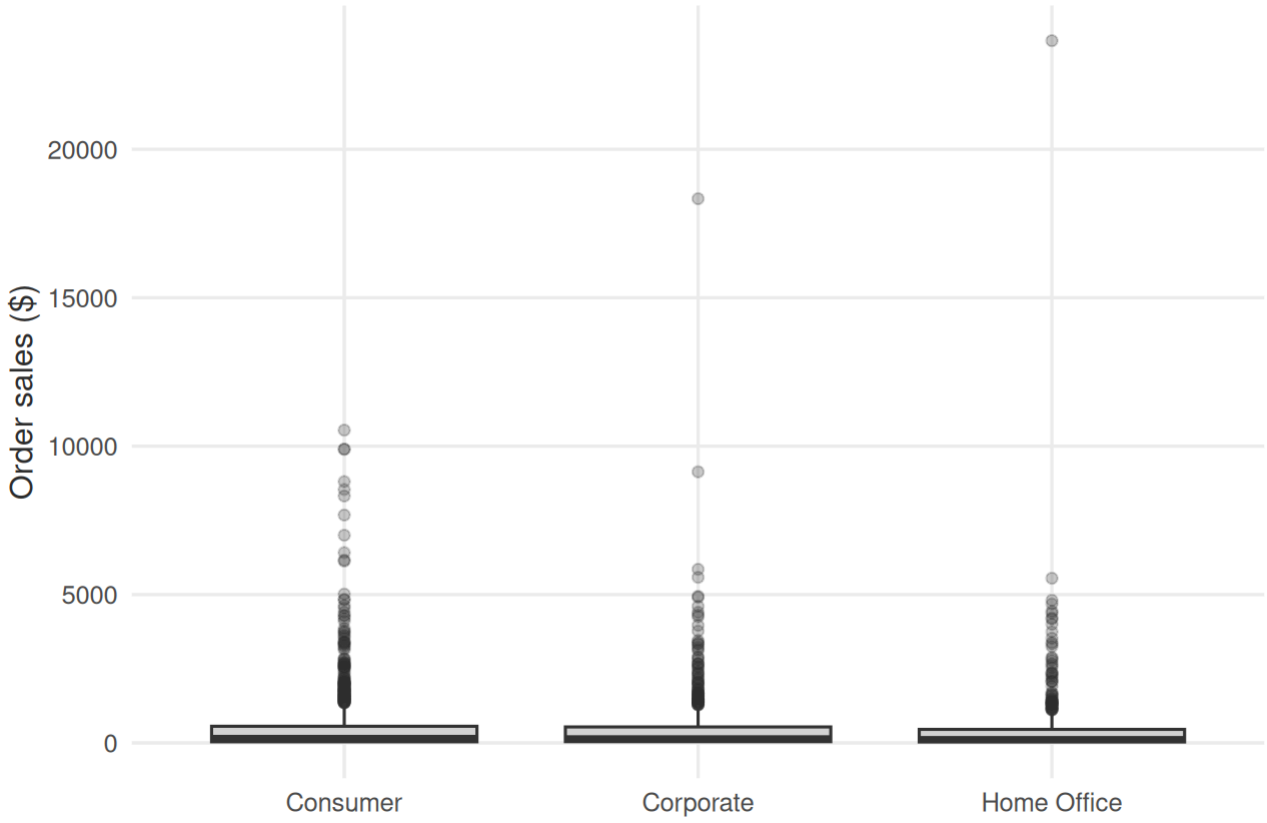
```
quantile(x, probs = c(0, .25, .5, .75, .9, .95, 1), type = 7)
```

```
      0%       25%       50%       75%       90%       95%      100%
  0.8360   38.7150  153.9820  535.3375 1190.8090 1799.6705 23661.2280
```

- A percentile describes a position in the ordered distribution.
- R's default `quantile(type = 7)` interpolates between observations.
- Different software conventions can give slightly different sample quartiles.

Boxplots and IQR fences

```
ggplot(orders, aes(segment, sales)) +
  geom_boxplot(fill = petra_accent, outlier.alpha = 0.25) +
  labs(x = NULL, y = "Order sales ($)")
```



Order sales by customer segment; points beyond the whiskers are flags, not deletion instructions.

```r
q <- quantile(x, c(.25, .75), type = 7)
fences <- c(lower = q[1] - 1.5 * IQR(x), upper = q[2] + 1.5 * IQR(x))
c(fences, flagged_orders = sum(x < fences[1] | x > fences[2]))
```

```
lower.25%      upper.75% flagged_orders
-706.2188      1280.2713       301.0000
```

- Box: middle 50%; internal line: median.
- Whiskers extend to the most extreme observations within the plotting fences.
- Boxplot hinges can differ from the reported type-7 quartiles for some sample sizes.
- Investigate flags for errors, genuine large transactions or a different business process.
- Retain valid extremes. If useful, report a clearly labelled sensitivity analysis alongside the full result.

Standardized values

```r
z <- (x - mean(x)) / sd(x)
c(min_z = min(z), max_z = max(z), beyond_3_sd = sum(abs(z) > 3))
```

```
     min_z       max_z beyond_3_sd
-0.4781004  23.5921053  53.0000000
```

- A z-score measures distance from the mean in SD units.
- The familiar 68–95–99.7 rule applies approximately to normal distributions.
- It is unsuitable as a blanket rule for skewed order values.

2.9 Put the evidence together

A useful management briefing contains:

1. **Scope:** US Superstore orders placed in 2023–2025; fictitious company.
2. **Finding:** Furniture has substantial sales but a thin recorded profit margin.
3. **Evidence:** category totals, weighted margins and the distribution of order values.
4. **Interpretation:** sales volume and profitability answer different questions.
5. **Next step:** examine subcategories, discount bands and large loss-making orders.
6. **Limit:** observed associations do not prove the effect of changing discounts.

Apply the methods

- Practical 1: Product mix and categorical tables.
- Practical 2: Order size and numerical measures.
- Practical 3: Discounts, profitability and group comparisons.
- Continue to Part 3 for uncertainty, tests and models.
- Optional transfer activity: Springville Herald, Section A applies descriptive methods to daily calls.

Sources

- Tableau Superstore sample data, downloaded 6 November 2026.
- Tableau on dimensions, measures and transaction detail.
- Course data dictionary and selection rules.
- Topic order builds on the instructor's original 307102 descriptive-statistics slides; business examples and explanations were revised for 307312 Applied Statistics.